

Exact Zero-Frequency Meissner Weight in a Gapless Frustration-Free QGN Parent

Graph-Hodge response, sharp Kohn–Drude limit criteria,
and protected multiblock stiffness

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Reviewer-revised research draft, July 27, 2026

Abstract

Finite-volume Peierls curvature is an adiabatic Kohn response, not automatically a bulk zero-frequency Meissner or superfluid weight. The thermodynamic, frequency, and transverse-momentum limits can fail to commute. We separate and solve these issues in a frustration-free paired lattice setting.

First, for an arbitrary analytic finite-volume ground cluster, the imaginary-frequency current kernel exceeds the Kohn curvature by an explicit positive spectral defect. After volume normalization, the thermodynamic-first Abelian kernel equals the thermodynamic Kohn curvature if and only if the current-generated spectral measure is uniformly tight at zero energy in the H^{-1} norm, assuming a uniform total H^{-1} bound. A uniform current-sector gap is sufficient but not necessary. For every positive-square parent $H(A) = D(A)^\dagger D(A)/2$, the entire finite-frequency response is the exact regularized least-squares residual

$$\mathcal{K}_L(i\zeta) = S_L^\dagger \frac{\zeta^2}{\mathcal{L}_L^2 + \zeta^2} S_L, \quad S_L = (\partial_A D_L)(0) P_L, \quad \mathcal{L}_L = \frac{1}{2} D_L D_L^\dagger.$$

Second, we close the independent transverse limit for an explicit local paired parent. On a connected graph, complete gauge-covariant site-swap rows turn the static least-squares subtraction into graph Hodge decomposition. In the fixed- n Dicke/AGP ground state,

$$\mathcal{C}_{G,n}[a] = j\gamma_{N,n} \langle a, P_{\ker B} a \rangle, \quad \mathcal{K}_{G,n}(i\zeta; a) = j\gamma_{N,n} \langle a, \frac{\zeta^2}{(jB^*B/4)^2 + \zeta^2} a \rangle, \quad \gamma_{N,n} = \frac{2n(N-n)}{N(N-1)},$$

where B is the oriented incidence matrix. Every lattice-transverse field satisfies $Ba = 0$, so its static and dynamical kernels coincide at every finite size, wave vector, and positive frequency regulator. On a periodic lattice in $d \geq 2$, with $n/N \rightarrow \rho \in (0, 1)$,

$$\boxed{\kappa_K = D_A = D_M = 2j\rho(1 - \rho) > 0}$$

in the electronic-link convention, or $D_s = j\rho(1 - \rho)/2$ in the pair-phase convention. The same state has exact pair off-diagonal long-range order, while its fixed-number gap closes as L^{-2} ; no uniform many-body gap is used. Finally, any multiblock positive-square extension containing the swap backbone inherits a frequency-independent positive transverse operator floor. The result is a genuine zero-temperature matter Meissner theorem for the exact parent. Transfer to finite-coupling microscopic downfoldings is deliberately left to a separate stability analysis.

Claim hierarchy.

1. The finite-volume Kubo–Kohn identity, the positive defect, the H^{-1} criterion, and the positive-square dynamical least-squares formula are exact finite-dimensional theorems.

2. Under a uniform total current H^{-1} bound, temporal Kohn–Drude equality is equivalent to uniform low-energy H^{-1} tightness. A closing complete many-body gap is allowed.
3. A Meissner statement requires an independent transverse limit. Temporal equality alone is not enough; a clean normal metal is a counterexample.
4. For the gauge-covariant swap-QGN/AGP parent, both temporal and transverse problems are solved algebraically at finite size. The resulting Meissner weight is positive at every interior filling.
5. Multiblock extensions inherit a protected transverse floor in every selected product composition. No scalar lowest-branch claim is made without a branch or state selector.
6. No finite-temperature theorem, unrestricted real-axis conductivity decomposition, material parameter extraction, or finite-coupling microscopic transfer theorem is claimed here.

1 Introduction

Exact paired parent Hamiltonians make static response unusually accessible. A complete Peierls twist defines an analytic finite-volume Hamiltonian; frustration freeness identifies the exact zero manifold; and second-order perturbation theory reduces the curvature to a positive least-squares residual. These ingredients underlie quantum-geometric-nesting (QGN) constructions, projected attractive-Hubbard models, and recent lower bounds on flat-band stiffness [16, 17]. They also meet the broader flat-band superfluid-weight literature, where quantum geometry supplies rigorous or mean-field bounds rather than the exact many-body response identity obtained below [8, 9].

The remaining distinction is physical rather than terminological. Kohn curvature is obtained by taking the frequency to zero at fixed finite volume and only then sending the volume to infinity [1]. A bulk uniform electric response takes the thermodynamic limit first. A Meissner response is a static transverse response at nonzero wave vector, followed by the transverse zero-momentum limit. The three limits diagnose different phenomena and are distinguished in the standard current-correlation criteria for an insulator, metal, or superconductor [2, 3, 4, 5]. Tada and Koma’s no-go analysis likewise shows why the order of limits and the class of applied fields must be specified in a superconductivity claim [7]. Explicit examples are known in which finite-size Kohn and thermodynamic Drude orders differ [6]; even when they agree, a clean normal metal can have Drude weight but no transverse Meissner response.

This paper gives a standalone resolution in two layers.

The first layer is general. We derive a positive operator measuring the difference between the finite-frequency kernel and static Kohn curvature. Its thermodynamic limit vanishes precisely when the current-generated low-energy spectral measure is uniformly integrable after weighting by inverse excitation energy. For positive-square parents, the same statement becomes a source-singular-mode criterion in the target space of the null map.

The second layer is model specific and exact. We introduce a gauge-covariant complete site-swap parent on a connected graph. Its seniority-zero compression is the symmetric-exclusion or ferromagnetic hard-core-boson generator associated with the Matsubara–Matsuda correspondence [12]. The current source is the graph divergence of the bond field. Static relaxation is therefore precisely Hodge projection onto gradients, while the residual is the graph cycle-space norm. Every divergence-free profile lies in the target kernel and has a frequency-independent response. On a periodic lattice this includes all lattice-transverse Fourier modes and the uniform harmonic flux.

The outcome is an exact gapless example with

$$\kappa_K = D_A = D_M > 0.$$

The paired Dicke state also has Yang off-diagonal long-range order [13], but the response identity is proved independently of that correlator. This distinction complements results deriving a Meissner effect from ODLRO under additional algebraic or thermodynamic hypotheses [10, 11]: here the finite-size current kernel itself is evaluated exactly before any thermodynamic limit. The state is also a zero-momentum

relative of the η -pairing family, whose ODLRO does not by itself settle the transport question [14]. The fixed-number spectral gap closes quadratically; the theorem works because gauge geometry separates transverse sources from the soft longitudinal sector, not because a hidden uniform gap survives.

2 Three responses and their limit orders

2.1 Gauge-covariant finite tori and profile-dependent kernels

Let Λ_L be a periodic d -dimensional lattice of volume V_L , and fix a conserved particle-number sector. Let $H_L[A]$ be a finite-range or uniformly exponentially local Hamiltonian coupled to a real bond vector potential by a complete Peierls prescription. Gauge covariance means

$$H_L[A + \nabla\phi] = U_\phi H_L[A] U_\phi^\dagger. \quad (1)$$

For any real bond profile a on Λ_L , set

$$H_L(t; a) = H_L[ta], \quad J_L[a] = \partial_t H_L(t; a)|_{t=0}, \quad T_L[a] = \partial_t^2 H_L(t; a)|_{t=0}. \quad (2)$$

A uniform flat connection in direction v is the harmonic profile $a_{L,v}^{\text{harm}}$; when no argument is displayed below, J_L , T_L , $\mathcal{C}_L$, and $\mathcal{K}_L$ refer to that profile.

Let E_L be the lowest zero-field energy, P_L the projector onto the complete finite-volume ground cluster, $Q_L = \mathbb{I} - P_L$, and

$$\hat{H}_L = H_L[0] - E_L \succeq 0. \quad (3)$$

The cluster is assumed isolated for each finite L , but no size-independent gap is assumed. For each profile used in a response coefficient we impose

$$P_L J_L[a] P_L = 0. \quad (4)$$

This holds automatically for positive-square parents and for time-reversal-symmetric zero-current branches. If it fails, first-order splitting or a cusp can precede any quadratic response.

2.2 Kohn, Abelian, and transverse Meissner kernels

For an arbitrary profile a , finite-dimensional degenerate perturbation theory gives the second-order Kohn operator

$$\mathcal{C}_L[a] = P_L T_L[a] P_L - 2P_L J_L[a] Q_L \hat{H}_L^+ Q_L J_L[a] P_L, \quad (5)$$

and, for $\zeta > 0$, the imaginary-frequency or Abel-regularized kernel

$$\mathcal{K}_L(i\zeta; a) = P_L T_L[a] P_L - 2P_L J_L[a] Q_L \frac{\hat{H}_L}{\hat{H}_L^2 + \zeta^2} Q_L J_L[a] P_L. \quad (6)$$

Write

$$c_L[a] = \frac{\mathcal{C}_L[a]}{V_L}, \quad k_L(\zeta; a) = \frac{\mathcal{K}_L(i\zeta; a)}{V_L}, \quad k_L(0; a) := c_L[a]. \quad (7)$$

The value at $\zeta = 0$ is therefore, by definition, the finite-volume static curvature; it is not a thermodynamic-first frequency limit.

For the harmonic uniform profile, abbreviate $c_L(v) = c_L[a_{L,v}^{\text{harm}}]$ and $k_L(\zeta; v) = k_L(\zeta; a_{L,v}^{\text{harm}})$. When the limits exist, define

$$\kappa_K(v) = \lim_{L \rightarrow \infty} c_L(v), \quad (8)$$

$$D_A(v) = \lim_{\zeta \downarrow 0} \lim_{L \rightarrow \infty} k_L(\zeta; v). \quad (9)$$

The first takes the static limit at finite volume; the second takes the thermodynamic limit first.

For $d \geq 2$, let $a_{L,q,v}$ be a real lattice-transverse bond profile with Fourier support at $\pm q$, polarization v , exact lattice divergence zero, and normalization $\|a_{L,q,v}\|^2 = V_L$. Define

$$\mathcal{K}_{L,T}(q, i\zeta; v) := \mathcal{K}_L(i\zeta; a_{L,q,v}), \quad k_{L,T}(q, \zeta; v) := \frac{\mathcal{K}_{L,T}(q, i\zeta; v)}{V_L}, \quad (10)$$

with

$$\mathcal{K}_{L,T}(q, 0; v) := \mathcal{C}_L[a_{L,q,v}], \quad k_{L,T}(q, 0; v) := c_L[a_{L,q,v}]. \quad (11)$$

Thus every occurrence of $k_{L,T}(q, 0; v)$ means that $\zeta \downarrow 0$ has already been taken at fixed finite L .

The allowed momenta depend on L . An admissible transverse zero-momentum limit is a sequence $q_m \rightarrow 0$ together with, for each m , a cofinal subsequence of tori on which $q_m \in \Lambda_L^*$ and the profiles $a_{L,q_m,v}$ are defined. For rational reciprocal vectors one may, for example, take L through integer multiples of a common period. The matter Meissner weight is

$$D_M(v) = \lim_{m \rightarrow \infty} \lim_{\substack{L \rightarrow \infty \\ q_m \in \Lambda_L^*}} k_{L,T}(q_m, 0; v), \quad (12)$$

when the result is independent of the admissible sequence. The alternative quantity

$$\lim_{m \rightarrow \infty} \lim_{\zeta \downarrow 0} \lim_{\substack{L \rightarrow \infty \\ q_m \in \Lambda_L^*}} k_{L,T}(q_m, \zeta; v) \quad (13)$$

is a separate order of limits unless temporal uniformity has also been established at nonzero q . In the exact swap model below the two definitions coincide already at finite size because the transverse kernel is independent of q and ζ .

The adjective “transverse” is lattice exact: the polarization is annihilated by the lattice divergence symbol, which approaches $q \cdot v = 0$ at small momentum.

Remark 2.1 (Matter response). The coefficient in (12) is the zero-temperature matter current kernel before coupling to the Maxwell field. Conventional factors of electric charge and optical π multiply all three coefficients and do not affect the order-of-limits statements.

3 Exact finite-volume Kubo–Kohn identity

Theorem 3.1 (Positive finite-frequency defect). *Under the hypotheses of Section 2, for every finite L , every profile a satisfying (4), and every $\zeta > 0$,*

$$\boxed{\mathcal{K}_L(i\zeta; a) - \mathcal{C}_L[a] = 2\zeta^2 P_L J_L[a] Q_L \hat{H}_L^+ (\hat{H}_L^2 + \zeta^2)^{-1} Q_L J_L[a] P_L \succeq 0.} \quad (14)$$

Consequently $\mathcal{K}_L(i\zeta; a)$ decreases in the Loewner order to $\mathcal{C}_L[a]$ as $\zeta \downarrow 0$.

Proof. Subtract (5) from (6). On the positive spectrum,

$$\frac{1}{E} - \frac{E}{E^2 + \zeta^2} = \frac{\zeta^2}{E(E^2 + \zeta^2)} \geq 0. \quad (15)$$

Functional calculus gives (14). The scalar multiplier is increasing in ζ , which gives monotonicity. $\square$

The theorem makes the finite-size statement exact. At fixed L , static curvature and the zero-frequency Kubo kernel coincide. The only temporal issue is uniformity in L .

3.1 Operator-valued current spectral measure

Let $P_{\hat{H}_L}(B)$ be the spectral projection of $\hat{H}_L$ for a Borel set $B \subset (0, \infty)$. Define the positive operator-valued measure on $P_L \mathcal{H}_L$ by

$$\Sigma_L(B) = \frac{1}{V_L} P_L J_L Q_L P_{\hat{H}_L}(B) Q_L J_L P_L. \quad (16)$$

Also put $d_L = V_L^{-1} P_L T_L P_L$. Then

$$c_L = d_L - 2 \int_{(0, \infty)} \frac{1}{E} d\Sigma_L(E), \quad (17)$$

$$k_L(\zeta) = d_L - 2 \int_{(0, \infty)} \frac{E}{E^2 + \zeta^2} d\Sigma_L(E), \quad (18)$$

$$k_L(\zeta) - c_L = 2 \int_{(0, \infty)} \frac{\zeta^2}{E(E^2 + \zeta^2)} d\Sigma_L(E). \quad (19)$$

At finite volume these are finite sums. State-resolved formulas follow by taking expectations in a selected ground-state density matrix.

Lemma 3.2 (The relevant gap is a current-sector gap). *Suppose $[H_L, N_L] = 0$ and the Peierls coupling preserves particle number. Then every matrix element in (16) lies in the same fixed-number sector as the ground cluster. An addition/removal charge gap therefore does not bound the denominators in (17). A uniform gap in the current-generated fixed-number subspace is sufficient.*

Proof. Particle-number conservation gives $[J_L, N_L] = 0$. Hence $J_L P_L$ lies in the same number sector as P_L , and only that sector enters the spectral measure. $\square$

4 Sharp temporal order-of-limits criterion

Definition 4.1 (H^{-1} boundedness and tightness). Define

$$M = \sup_L \left\| \int_{(0, \infty)} E^{-1} d\Sigma_L(E) \right\|. \quad (20)$$

The current measures are uniformly H^{-1} -tight at zero (equivalently, uniformly H^{-1} equi-integrable at zero energy) if

$$\lim_{\delta \downarrow 0} \limsup_{L \rightarrow \infty} \left\| \int_{(0, \delta]} E^{-1} d\Sigma_L(E) \right\| = 0. \quad (21)$$

Theorem 4.2 (Sharp temporal criterion). *Assume $M < \infty$. Then the following are equivalent:*

- (i) *the current measures satisfy (21);*
- (ii) *the finite-frequency defect vanishes uniformly in the thermodynamic limit,*

$$\lim_{\zeta \downarrow 0} \limsup_{L \rightarrow \infty} \|k_L(\zeta) - c_L\| = 0. \quad (22)$$

More quantitatively, put

$$t(\delta) = \limsup_{L \rightarrow \infty} \left\| \int_{(0, \delta]} E^{-1} d\Sigma_L(E) \right\|. \quad (23)$$

Then

$$t(\zeta) \leq \limsup_{L \rightarrow \infty} \|k_L(\zeta) - c_L\|, \quad (24)$$

and, for $0 < \zeta < \delta$,

$$\limsup_{L \rightarrow \infty} \|k_L(\zeta) - c_L\| \leq 2t(\delta) + 2\frac{\zeta^2}{\delta^2}M. \quad (25)$$

Proof. For $0 < E \leq \zeta$,

$$2\frac{\zeta^2}{E(E^2 + \zeta^2)} \geq \frac{1}{E}, \quad (26)$$

so positivity of the operator-valued measure gives (24). To prove (25), split (19) at δ . Below δ , the scalar kernel is at most $2/E$. Above δ , it is at most $2(\zeta^2/\delta^2)E^{-1}$. The triangle inequality for positive operator integrals yields the bound. The equivalence follows by first taking $L \rightarrow \infty$, then $\zeta \downarrow 0$, and in the upper bound finally $\delta \downarrow 0$. $\square$

Corollary 4.3 (Kohn equals the bulk Abelian kernel). *Assume Theorem 4.2, assume $c_L \rightarrow \kappa_K$ in operator norm, and assume $k_L(\zeta)$ has a thermodynamic limit for every fixed $\zeta > 0$. Then*

$$\boxed{D_A = \kappa_K}. \quad (27)$$

The same statement holds after taking an expectation in any consistently selected sequence of states or density matrices in the ground cluster.

Corollary 4.4 (Uniform current-sector gap). *If the spectral support of $Q_L J_L P_L$ lies in $[\Delta, \infty)$ with $\Delta > 0$ independent of L , and if (20) holds, then (21) holds. The converse is false: the full gap may close while the current matrix elements vanish sufficiently rapidly.*

Remark 4.5 (Mode-by-mode diagnostic). For a selected scalar ground state, let a low mode have energy Δ_L and normalized current weight

$$w_L = \frac{1}{V_L} |\langle m_L | J_L | \Omega_L \rangle|^2. \quad (28)$$

Its contribution to the tightness functional is w_L/Δ_L . The mode is harmless if this ratio tends to zero and obstructing if it has a positive limit. The theorem is therefore a matrix-element statement, not merely a dispersion statement.

5 Positive-square parents: dynamical least squares

Let

$$H_L(a) = \frac{1}{2} D_L(a)^\dagger D_L(a), \quad D_L(a) : \mathcal{H}_L \rightarrow \mathcal{T}_L \quad (29)$$

be analytic, and let $P_L = P_{\ker D_L(0)}$. Positivity gives $E_L = 0$, so throughout this section $\hat{H}_L = H_L(0)$. Put

$$D_L = D_L(0), \quad S_L = (\partial_a D_L)(0)P_L, \quad \mathcal{L}_L = \frac{1}{2} D_L D_L^\dagger. \quad (30)$$

The map S_L is the stacked Peierls source and $\mathcal{L}_L$ is the target-space Laplacian.

Theorem 5.1 (Dynamical least-squares identity). *For every finite L and $\zeta > 0$,*

$$\boxed{\mathcal{K}_L(i\zeta) = S_L^\dagger \frac{\zeta^2}{\mathcal{L}_L^2 + \zeta^2} S_L}. \quad (31)$$

Its static Kohn curvature is

$$\boxed{C_L = S_L^\dagger P_{\ker D_L^\dagger} S_L.} \quad (32)$$

Equivalently, for $\psi \in P_L \mathcal{H}_L$,

$$\langle \psi | \mathcal{K}_L(i\zeta) | \psi \rangle = \min_{y \in \mathcal{T}_L} \left\{ \|S_L \psi - \mathcal{L}_L y\|^2 + \zeta^2 \|y\|^2 \right\}. \quad (33)$$

Proof. Since $D_L P_L = 0$,

$$J_L P_L = \frac{1}{2} D_L^\dagger S_L, \quad P_L T_L P_L = S_L^\dagger S_L. \quad (34)$$

Singular-value functional calculus gives

$$\frac{1}{2} D_L \frac{\widehat{H}_L}{\widehat{H}_L^2 + \zeta^2} D_L^\dagger = \frac{\mathcal{L}_L^2}{\mathcal{L}_L^2 + \zeta^2}. \quad (35)$$

Substitution into (6) proves (31). Taking $\zeta \downarrow 0$ projects onto $\ker \mathcal{L}_L = \ker D_L^\dagger$ and gives (32). Minimizing the quadratic form in (33) gives the normal equation $(\mathcal{L}_L^2 + \zeta^2)y = \mathcal{L}_L S_L \psi$ and reproduces (31). $\square$

Corollary 5.2 (Sharp source-singular-mode criterion). *Assume*

$$\sup_L \frac{1}{V_L} \|S_L^\dagger S_L\| < \infty. \quad (36)$$

Then temporal Kohn–Drude equality holds if and only if

$$\lim_{\delta \downarrow 0} \limsup_{L \rightarrow \infty} \frac{1}{V_L} \|S_L^\dagger \mathbf{1}_{(0, \delta]}(\mathcal{L}_L) S_L\| = 0. \quad (37)$$

Thus a closing positive singular-value sector is harmless precisely when it carries vanishing extensive Peierls-source weight.

Proof. On the positive singular spectrum, D_L identifies the state-space current measure with one half of the target-space source measure. Equivalently, subtract (32) from (31); the defect multiplier is $\zeta^2/(\lambda^2 + \zeta^2)$ and the same cutoff argument as in Theorem 4.2 applies. $\square$

6 From a uniform kernel to a transverse Meissner weight

Definition 6.1 (Static transverse continuity). Choose a sequence of ground states or density matrices defining the thermodynamic phase. The static transverse response is continuous at zero momentum in direction v if, for every admissible sequence specified after (11),

$$\lim_{m \rightarrow \infty} \lim_{\substack{L \rightarrow \infty \\ q_m \in \Lambda_L^*}} k_{L,T}(q_m, 0; v) = \lim_{L \rightarrow \infty} c_L(v). \quad (38)$$

Theorem 6.2 (Kohn–Drude–Meissner completion). *Suppose, in a selected thermodynamic state,*

- (i) *the hypotheses of Theorem 4.3 hold for the uniform current;*
- (ii) *static transverse continuity (38) holds.*

Then

$$\boxed{\kappa_K = D_A = D_M.} \quad (39)$$

Proof. The first equality is Theorem 4.3; the second is (38) and the definition (12). $\square$

The theorem is modular. Temporal failure is low-energy inverse-energy current weight. Spatial failure is a discontinuity between an exactly uniform connection and the limit of nonzero transverse fields. The two obstructions are independent.

Proposition 6.3 (A sufficient summability condition). *Let $j_v(x)$ be a local current density in a translation-invariant selected pure ground state and write $J_{L,v} = \sum_x j_v(x)$. Define the connected Euclidean correlation by*

$$G_{L,v}(x, t) = \langle \Omega_L | j_v(x) e^{-t\hat{H}_L} Q_L j_v(0) | \Omega_L \rangle, \quad t \geq 0. \quad (40)$$

The insertion of Q_L is the connected subtraction. Assume local thermodynamic limits exist, translation invariance holds along the chosen sequence, and there is an L -independent nonnegative envelope F such that

$$|G_{L,v}(x, t)| \leq F(x, t), \quad \sum_x \int_0^\infty F(x, t) dt < \infty. \quad (41)$$

Assume also that the complete finite- q diamagnetic term has a thermodynamic limit continuous at $q = 0$. Then the uniform current is H^{-1} -tight and the static transverse kernel is continuous at $q = 0$. The same conclusion holds for a finite ground multiplet if the bound is uniform for all matrix elements in the selected density matrix.

Proof. Let $\mu_{L,v}$ be the scalar volume-normalized spectral measure of $J_{L,v}$ in the selected state. Translation invariance and the spectral theorem give

$$g_{L,v}(t) := \frac{1}{V_L} \langle \Omega_L | J_{L,v} e^{-t\hat{H}_L} Q_L J_{L,v} | \Omega_L \rangle = \sum_x G_{L,v}(x, t) = \int_{(0,\infty)} e^{-tE} d\mu_{L,v}(E). \quad (42)$$

The connected subtraction is precisely the projector Q_L , so no zero-energy atom occurs. The summable envelope implies $|g_{L,v}(t)| \leq \sum_x F(x, t)$ with an L -independent integrable majorant. Moreover, for $0 < E \leq \delta$,

$$\frac{1}{E} \leq e \int_{1/\delta}^\infty e^{-tE} dt. \quad (43)$$

Therefore positivity and (42) imply

$$\int_{(0,\delta]} \frac{d\mu_{L,v}(E)}{E} \leq e \int_{1/\delta}^\infty g_{L,v}(t) dt \leq e \sum_x \int_{1/\delta}^\infty F(x, t) dt. \quad (44)$$

The right side tends to zero uniformly in L , proving the H^{-1} tightness criterion.

For the static nonzero-momentum paramagnetic susceptibility, the same spectral representation gives

$$\chi_{L,v}(q) = \sum_x e^{-iq \cdot x} \int_0^\infty G_{L,v}(x, t) dt. \quad (45)$$

Local thermodynamic convergence and (41) permit the $L \rightarrow \infty$ limit term by term; absolute summability then permits $q \rightarrow 0$ by dominated convergence. Hence the thermodynamic paramagnetic term is continuous at zero momentum. Adding the assumed continuous diamagnetic term proves (38). $\square$

The exact model below uses a stronger mechanism: a finite-size source-kernel identity removes the need for asymptotic correlation estimates.

7 Gauge-covariant swap-QGN parent on a graph

7.1 Fermionic definition

Let $G = (V, E)$ be a finite connected graph with $N = |V|$ vertices and a fixed orientation of each edge. At site x place spinful fermions $c_{x\sigma}$, $\sigma = \uparrow, \downarrow$, with

$$N_x = n_{x\uparrow} + n_{x\downarrow}, \quad p_x^\dagger = c_{x\uparrow}^\dagger c_{x\downarrow}^\dagger, \quad q_x = (n_{x\uparrow} - n_{x\downarrow})^2. \quad (46)$$

Let W_e be the complete fermionic site swap on the endpoints of $e = (x \rightarrow y)$, so that it exchanges all local creation operators at x and y and leaves the other sites fixed. Define the Peierls-covariant swap and projector row

$$W_e[A] = e^{-iA_e N_y} W_e e^{iA_e N_y}, \quad \Pi_e[A] = \frac{\mathbb{I} - W_e[A]}{2}. \quad (47)$$

Since $W_e[A]$ is a Hermitian involution, $\Pi_e[A]$ is a projector. The parent is

$$H_G[A] = \frac{U}{2} \sum_{x \in V} q_x^2 + \frac{j}{2} \sum_{e \in E} \Pi_e[A]^2, \quad U, j > 0. \quad (48)$$

It is a local positive-square Hamiltonian.

Let $B : \mathbb{R}^E \rightarrow \mathbb{R}^V$ be the oriented incidence matrix,

$$B_{ze} = \begin{cases} -1, & z = t(e), \\ +1, & z = h(e), \\ 0, & \text{otherwise.} \end{cases} \quad (49)$$

For a site function ϕ , $B^* \phi$ is the discrete gradient.

Proposition 7.1 (Complete gauge covariance). *With*

$$U_\phi = \exp\left(-i \sum_x \phi_x N_x\right), \quad (50)$$

one has

$$U_\phi H_G[A] U_\phi^\dagger = H_G[A + B^* \phi]. \quad (51)$$

Proof. For $e = (x \rightarrow y)$, conjugation by U_ϕ multiplies a fermion transferred from x to y by $e^{-i(\phi_y - \phi_x)}$ and the reverse transfer by the conjugate phase. This is exactly the replacement $A_e \mapsto A_e + (B^* \phi)_e$ in (47). The onsite rows are gauge invariant. $\square$

7.2 Exact fixed-number ground state

For $0 \leq n \leq N$, define the normalized paired Dicke/AGP state

$$|\Omega_{N,n}\rangle = \binom{N}{n}^{-1/2} \sum_{\substack{S \subseteq V \\ |S|=n}} \prod_{x \in S} p_x^\dagger |0\rangle. \quad (52)$$

Theorem 7.2 (Complete fixed- $2n$ zero space). *At $A = 0$, in the sector of total particle number $2n$,*

$$\ker H_G[0]|_{2n} = \text{span}\{|\Omega_{N,n}\rangle\}. \quad (53)$$

Odd-particle sectors have no zero vectors.

Proof. The rows q_x force each site to be empty or doubly occupied. In that seniority-zero subspace, each W_e is the ordinary transposition of the hard-core pair occupations at the endpoints. The rows Π_e require invariance under every edge transposition. Because G is connected, the edge transpositions generate the full symmetric group on V , whose fixed subspace at pair number n is the one-dimensional uniform subset state (52). A state with odd total particle number contains a singly occupied site and is penalized by an onsite row. $\square$

7.3 Seniority-zero compression and one-body functions

Write a hard-core pair configuration as $|S\rangle$, $S \subset V$, $|S| = n$. In the paired sector,

$$H_{G,n} = \frac{j}{4} \sum_{e \in E} (\mathbb{I} - W_e). \quad (54)$$

With $m_x = S_x^z + \frac{1}{2}$ and $p_x^\dagger = S_x^+$, this is exactly the isotropic spin- $\frac{1}{2}$ ferromagnet

$$H_{G,n} = \frac{j}{2} \sum_{e=(x,y)} \left(\frac{1}{4} - \mathbf{S}_x \cdot \mathbf{S}_y \right), \quad (55)$$

restricted to fixed magnetization. The Peierls field twists the XY exchange with charge two. This form makes both the quadratic magnon gap and the graph-Hodge separation of longitudinal and transverse sources transparent. The onsite coefficient $U > 0$ is used only to select seniority zero; it drops out of all response formulas inside the paired ground sector. For a real site function f with $\sum_x f_x = 0$, define

$$|f\rangle = \binom{N}{n}^{-1/2} \sum_{|S|=n} \left(\sum_{x \in S} f_x \right) |S\rangle. \quad (56)$$

Let

$$\alpha_{N,n} = \frac{n(N-n)}{N(N-1)}, \quad \gamma_{N,n} = 2\alpha_{N,n}. \quad (57)$$

Lemma 7.3 (Exact site-function representation). *For zero-mean f, g ,*

$$\langle f|g\rangle = \alpha_{N,n} \langle f, g\rangle, \quad (58)$$

$$H_{G,n}|f\rangle = \frac{j}{4} |BB^*f\rangle. \quad (59)$$

Proof. For (58), the probability that a uniformly chosen n -subset contains x is n/N , and the probability that it contains distinct x, y is $n(n-1)/[N(N-1)]$. Expanding the two subset sums and using $\sum_x f_x = \sum_x g_x = 0$ leaves the coefficient $\alpha_{N,n}$.

For (59), an edge transposition changes $\sum_{z \in S} f_z$ by the endpoint difference whenever exactly one endpoint is occupied. Summing over edges gives the graph Laplacian BB^* acting on f . $\square$

8 Exact graph-Hodge response

Let $a \in \mathbb{R}^E$ be a real bond profile and set $A = ta$. Define

$$J_G[a] = \partial_t H_G[ta]|_{t=0}, \quad T_G[a] = \partial_t^2 H_G[ta]|_{t=0}. \quad (60)$$

The graph Hodge projector onto the cycle space is

$$P_{\text{cyc}} = P_{\ker B} = \mathbb{I} - B^*(BB^*)^+ B. \quad (61)$$

Lemma 8.1 (Exact current and stress sources). *On the fixed- n ground state,*

$$J_G[a]|\Omega_{N,n}\rangle = \frac{ij}{2}|Ba\rangle, \quad (62)$$

$$\langle\Omega_{N,n}|T_G[a]|\Omega_{N,n}\rangle = j\gamma_{N,n}\|a\|^2. \quad (63)$$

Proof. On an edge $e = (x \rightarrow y)$, the twisted swap multiplies a pair moved from x to y by e^{-2ita_e} and the reverse move by e^{2ita_e} . Differentiating the uniform subset state therefore produces the signed endpoint occupation difference. Summing over edges gives the divergence Ba and the coefficient in (62). A second derivative contributes only when the edge has exactly one occupied endpoint. That probability is $2n(N-n)/[N(N-1)] = \gamma_{N,n}$, proving (63). $\square$

Theorem 8.2 (Exact graph-Hodge static and dynamical kernels). *For every finite connected graph, every pair number $0 \leq n \leq N$, every real bond profile a , and every $\zeta > 0$,*

$$\boxed{\mathcal{C}_{G,n}[a] = j\gamma_{N,n}\langle a, P_{\text{cyc}}a \rangle}, \quad (64)$$

$$\boxed{\mathcal{K}_{G,n}(i\zeta; a) = j\gamma_{N,n}\langle a, \frac{\zeta^2}{(jB^*B/4)^2 + \zeta^2}a \rangle}. \quad (65)$$

Consequently:

(i) *if $a = B^*\phi$ is a gauge gradient, then $\mathcal{C}_{G,n}[a] = 0$;*

(ii) *if $Ba = 0$, then*

$$\mathcal{K}_{G,n}(i\zeta; a) = \mathcal{C}_{G,n}[a] = j\gamma_{N,n}\|a\|^2 \quad (66)$$

for every $\zeta > 0$.

Proof. By Theorem 8.1, the current source lies in the site-function sector. On that sector, Theorem 7.3 identifies the Hamiltonian with $(j/4)BB^*$. Hence the paramagnetic term in (6) is

$$\frac{j^2}{2}\alpha_{N,n}\langle Ba, \frac{(j/4)BB^*}{[(j/4)BB^*]^2 + \zeta^2}Ba \rangle. \quad (67)$$

Using the common nonzero spectrum of BB^* and B^*B , and subtracting this term from (63), gives (65). Taking $\zeta \downarrow 0$ projects onto $\ker(B^*B) = \ker B$ and proves (64). The two consequences follow from the orthogonal Hodge decomposition $\mathbb{R}^E = \text{Ran } B^* \oplus \ker B$. $\square$

Remark 8.3 (Exact finite-frequency graph filter). Equation (65) is the model's central infrared mechanism. On an edge-Laplacian eigenmode with eigenvalue $\lambda > 0$, the dynamical multiplier is

$$\frac{\zeta^2}{(j\lambda/4)^2 + \zeta^2}. \quad (68)$$

At fixed ζ , sufficiently soft longitudinal modes remain unresolved; the static limit removes them. Cycle-space modes have $\lambda = 0$ and survive with multiplier one at every frequency. The general H^{-1} criterion is therefore realized here as an exact graph filter, without a uniform many-body gap. For a divergence-free field, (62) gives $J_G[a]|\Omega_{N,n}\rangle = 0$ already at finite size. The transverse response is consequently purely diamagnetic, independent of q , and independent of ζ . This is a deliberately fine-tuned feature of the frustration-free parent, stronger than the partial diamagnetic-paramagnetic cancellation in a generic BCS state.

9 Zero-frequency Meissner theorem on periodic lattices

Let G_L be the nearest-neighbor graph of a periodic d -dimensional Bravais lattice with $N = L^d$ sites. A Fourier bond field $a_\mu(q)$ is lattice transverse when

$$\sum_{\mu=1}^d (1 - e^{-iq_\mu}) a_\mu(q) = 0. \quad (69)$$

For every allowed nonzero q in $d \geq 2$, there is a nonzero transverse polarization. Normalize the corresponding real profile by $\|a\|^2 = N$.

Theorem 9.1 (Exact zero-frequency Meissner weight). *Let $d \geq 2$, let $n_L/N \rightarrow \rho \in (0, 1)$, and consider the parent (48) on G_L . Then every finite-volume uniform harmonic profile, every allowed lattice-transverse Fourier profile, and every $\zeta > 0$ have the same response density,*

$$\frac{1}{N} \mathcal{K}_{G_L, n_L}(i\zeta; a) = \frac{1}{N} \mathcal{C}_{G_L, n_L}[a] = j\gamma_{N, n_L}. \quad (70)$$

All admissible orders of $L \rightarrow \infty$, $\zeta \downarrow 0$, and transverse $q \rightarrow 0$ therefore agree:

$$\boxed{\kappa_K = D_A = D_M = 2j\rho(1 - \rho) > 0.} \quad (71)$$

If the pair phase is $Q = 2A$, the corresponding pair-phase stiffness is

$$\boxed{D_s = \frac{j}{2}\rho(1 - \rho).} \quad (72)$$

Proof. A constant directional bond profile on a torus is divergence free, as is every field satisfying (69). Equation (66) and the normalization $\|a\|^2 = N$ give (70). Since

$$\gamma_{N, n_L} = \frac{2n_L(N - n_L)}{N(N - 1)} \longrightarrow 2\rho(1 - \rho), \quad (73)$$

all limit orders give (71). Differentiating with respect to $Q = 2A$ instead of A multiplies the curvature by 1/4 and gives (72). $\square$

9.1 Pair ODLRO and the closing fixed-number gap

Proposition 9.2 (Exact pair-density matrix). *In the state (52),*

$$\langle \Omega_{N, n} | p_x^\dagger p_y | \Omega_{N, n} \rangle = \begin{cases} n/N, & x = y, \\ \alpha_{N, n}, & x \neq y. \end{cases} \quad (74)$$

The largest eigenvalue of this matrix is

$$\lambda_{\max} = \frac{n(N - n + 1)}{N}, \quad (75)$$

which is extensive at fixed interior density. The finite-volume electronic-link Meissner kernel satisfies

$$j\gamma_{N, n} = 2j \langle \Omega_{N, n} | p_x^\dagger p_y | \Omega_{N, n} \rangle \quad (x \neq y). \quad (76)$$

Proof. The diagonal entry is the probability that a uniformly chosen n -subset contains x . For $x \neq y$, $p_x^\dagger p_y$ maps a subset containing y but not x to another subset with the same uniform amplitude. The number of such subsets is $\binom{N-2}{n-1}$, giving $\alpha_{N, n}$. A matrix with constant diagonal and constant off-diagonal entries has uniform eigenvalue $n/N + (N - 1)\alpha_{N, n}$, which is (75). Equation (76) follows from (57). $\square$

Proposition 9.3 (No hidden uniform gap). *In the fixed- n paired sector on a periodic nearest-neighbor lattice,*

$$\text{gap}(H_{G_L,n}) = \frac{j}{4} \lambda_1(BB^*) = j \sin^2\left(\frac{\pi}{L}\right) = O(L^{-2}), \quad (77)$$

for $1 \leq n \leq N-1$. Here λ_1 is the smallest nonzero graph-Laplacian eigenvalue.

Proof. The upper bound and eigenvalue are realized by the site-function vectors in Theorem 7.3. Equality of the symmetric-exclusion spectral gap with the one-particle random-walk gap is the Aldous spectral-gap theorem [15]. For the periodic lattice, $\lambda_1(BB^*) = 4 \sin^2(\pi/L)$. $\square$

The Meissner theorem is therefore not a uniform-gap corollary. The gapless modes are longitudinal site functions. A transverse bond field has $Ba = 0$, so its physical current source vanishes before any pseudoinverse is applied.

10 Protected multiblock stiffness floor

Consider r paired blocks on the same graph. Let the full positive-square target map be stacked as

$$D[A] = D_{\text{sw}}[A] \oplus D_{\text{r}}[A], \quad (78)$$

where D_{sw} contains a complete swap row for every graph edge in each block, with coupling j_b , and every row of D_{r} annihilates the product-Dicke zero manifold at $A = 0$. Let S_{sw} and S_{r} be the corresponding Peierls source maps.

Theorem 10.1 (Source-kernel backbone protection). *For every divergence-free profile $Ba = 0$ and every $\zeta > 0$,*

$$\mathcal{K}(\text{i}\zeta; a) \succeq Q_{\text{sw}}[a], \quad (79)$$

$$\mathcal{C}[a] \succeq Q_{\text{sw}}[a], \quad (80)$$

where, in the product-composition basis $|\mathbf{n}\rangle$,

$$Q_{\text{sw}}[a] = \|a\|^2 \sum_{\mathbf{n}} \left(\sum_{b=1}^r j_b \gamma_{N,n_b} \right) |\mathbf{n}\rangle \langle \mathbf{n}|. \quad (81)$$

Thus additional positive-square multiplicative, remote-control, or interaction rows cannot cancel the transverse stiffness carried by the swap backbone.

Proof. For $Ba = 0$, the single-block theorem gives $D_{\text{sw}}^\dagger S_{\text{sw}} = 0$. Because the target labels are a direct sum, $s_0 := (S_{\text{sw}}, 0) \in \ker D^\dagger = \ker \mathcal{L}$ for the complete map. Hence $f_\zeta(\mathcal{L})s_0 = s_0$ for $f_\zeta(\mathcal{L}) = \zeta^2(\mathcal{L}^2 + \zeta^2)^{-1}$. Although $\mathcal{L} = DD^\dagger/2$ need not preserve the swap/residual block decomposition, self-adjointness now gives, for $s_{\text{r}} := (0, S_{\text{r}})$,

$$\langle s_0, f_\zeta(\mathcal{L})s_{\text{r}} \rangle = \langle f_\zeta(\mathcal{L})s_0, s_{\text{r}} \rangle = \langle s_0, s_{\text{r}} \rangle = 0. \quad (82)$$

Thus it is the kernel eigenvector property, followed by direct-sum orthogonality, that removes the cross terms. Applying Theorem 5.1 gives the direct positive swap Gram matrix plus a positive residual term. The static statement follows identically with $P_{\ker D^\dagger}$ in place of $f_\zeta(\mathcal{L})$. Different blocks preserve separate block charges and occupy orthogonal row labels, yielding (81). $\square$

Corollary 10.2 (Selected interior compositions are Meissner active). *On a periodic $d \geq 2$ lattice, choose a product composition with $n_{b,L}/N \rightarrow \rho_b$. Then*

$$D_{\text{M}} \geq 2 \sum_{b=1}^r j_b \rho_b (1 - \rho_b), \quad (83)$$

and the same lower bound holds for the Abelian zero-frequency kernel. It is strictly positive whenever at least one block is neither empty nor saturated.

Remark 10.3 (Ground-manifold selection). In an unresolved composition manifold, the lowest eigenvalue of the full curvature operator need not equal its diagonal expectation in a chosen product composition. A scalar physical response therefore requires a branch selector, a prepared coherent state, or a density matrix. The operator inequalities (79)–(80) remain valid without choosing one.

11 Independent failure mechanisms

11.1 Temporal failure from a soft positive target mode

Proposition 11.1 (Positive-square order-of-limits defect). *Let $V_L \rightarrow \infty$, $\Delta_L \downarrow 0$, and let the state space be spanned by $|g\rangle, |e\rangle$ and the target space by $|r\rangle, |z\rangle$. Define*

$$D_L(a) = \sqrt{2\Delta_L}|r\rangle\langle e| + a\left(\sqrt{2\alpha V_L}|r\rangle + \sqrt{c_0 V_L}|z\rangle\right)\langle g|, \quad \alpha > 0, \quad c_0 \geq 0, \quad (84)$$

and $H_L(a) = D_L(a)^\dagger D_L(a)/2$. At $a = 0$, $|g\rangle$ is the unique zero state and $|e\rangle$ has energy Δ_L . Nevertheless,

$$\frac{1}{V_L}\mathcal{C}_L = c_0, \quad (85)$$

$$\frac{1}{V_L}\mathcal{K}_L(i\zeta) = c_0 + 2\alpha \frac{\zeta^2}{\Delta_L^2 + \zeta^2}. \quad (86)$$

Hence

$$\lim_{L \rightarrow \infty} \frac{\mathcal{C}_L}{V_L} = c_0, \quad \lim_{\zeta \downarrow 0} \lim_{L \rightarrow \infty} \frac{\mathcal{K}_L(i\zeta)}{V_L} = c_0 + 2\alpha. \quad (87)$$

Proof. At zero twist, $\mathcal{L}_L = \Delta_L|r\rangle\langle r|$ and the source is the second term in (84) acting on $|g\rangle$. The static projector onto $\ker \mathcal{L}_L = \text{span}\{|z\rangle\}$ gives (85); Theorem 5.1 gives (86). Taking the limits proves (87). The soft mode carries source weight $2\alpha V_L$ at target energy Δ_L , so (37) fails exactly. $\square$

11.2 Spatial failure: Drude without Meissner

Proposition 11.2 (Clean normal metal). *Consider continuum spinless fermions on a periodic box,*

$$H_L(A) = \sum_{a=1}^N \frac{(p_a - A)^2}{2m}, \quad (88)$$

with a closed-shell ground state and density $n = N/V_L$. The finite-volume Kohn curvature and uniform Drude kernel are n/m . The static transverse kernel instead vanishes in the thermodynamic limit:

$$D_M = 0, \quad \kappa_K = D_A = \frac{n}{m}. \quad (89)$$

Proof. Along a fixed zero-temperature closed-shell adiabatic branch, differentiating (88) twice gives n/m . To evaluate the transverse thermodynamic limit without distributional shorthand, first use the smooth Fermi occupation $n_\beta(\epsilon) = [1 + e^{\beta(\epsilon - \mu)}]^{-1}$ at finite β . For q along x and polarization along y ,

$$\Pi_{yy}^{(\beta)}(q, 0) = \int \frac{d^d k}{(2\pi)^d} \left(\frac{k_y}{m}\right)^2 \frac{n_\beta(\epsilon_{k+q/2}) - n_\beta(\epsilon_{k-q/2})}{\epsilon_{k+q/2} - \epsilon_{k-q/2}}. \quad (90)$$

At fixed β , dominated convergence as $q \rightarrow 0$ gives the derivative $n'_\beta(\epsilon_k)$. The identity

$$\partial_{k_y} \left(\frac{k_y}{m} n_\beta(\epsilon_k) \right) = \frac{1}{m} n_\beta(\epsilon_k) + \frac{k_y^2}{m^2} n'_\beta(\epsilon_k) \quad (91)$$

and integration by parts yield $\lim_{q \rightarrow 0} \Pi_{yy}^{(\beta)}(q, 0) = -n_\beta/m$, where n_β is the particle density at that regularization. Taking $\beta \rightarrow \infty$ at fixed density gives $-n/m$, equivalently the usual zero-temperature Fermi-surface distribution. It cancels the diamagnetic term. The finite- β numerical check in the reproducibility suite tests this smooth regularization; the proposition is its zero-temperature limit. The spatial continuity hypothesis of Theorem 6.2 therefore fails. $\square$

These examples show why both theorem layers are necessary. The first violates temporal tightness in a frustration-free positive-square family. The second satisfies the uniform electric response but violates the transverse limit.

12 Theorem boundary and relation to microscopic transfer

The exact parent has now crossed the conceptual boundary between static Peierls curvature and a genuine zero-frequency transverse matter response. The proof uses five pieces that can be audited separately:

Question	Sufficient certificate	Failure signature
Finite-volume branch	isolated ground cluster and $PJP = 0$	linear splitting, cusp, or unresolved selector
Complete gauge coupling	bondwise covariance and both current/stress derivatives	partial twist or missing diamagnetic term
Temporal uniformity	current H^{-1} tightness, equivalently source tightness	finite inverse-energy weight in soft modes
Transverse completion	static $q \rightarrow 0$ continuity or exact source-kernel identity	Drude response with transverse cancellation
Exact swap route	$Ba = 0$ and $S[a] \subset \ker D^\dagger$	source outside the target kernel

The finite-coupling microscopic question is different. A local Schrieffer–Wolff low block can reproduce the parent at leading order while carrying a quasi-local higher-order remainder. Because the parent gap closes as L^{-2} , a small local operator norm does not by itself imply a small response error: the remainder can move the selected state or couple a transverse source to a soft longitudinal mode. The required transfer theorem must control both state and current sources in a relative H^{-1} topology, uniformly in the transverse infrared window. That analysis is intentionally separated from the exact Meissner theorem proved here.

13 Conclusion

Static Peierls curvature becomes a dynamical statement only after an infrared uniformity theorem. The sharp temporal condition is not that the complete many-body gap stay open, but that the physical current carry no residual inverse-energy weight into asymptotically soft excitations. In a positive-square parent, this is exactly a statement about the projection of the Peierls source onto soft positive target singular modes.

A Meissner statement additionally requires the transverse spatial limit. The graph swap parent solves that problem at finite size. Its current source is the graph divergence of the bond field; least-squares

relaxation is Hodge projection; and every divergence-free field lies exactly in the target kernel. The transverse kernel is therefore independent of frequency and wave vector before any thermodynamic limit is taken.

The resulting periodic-lattice model has

$$\kappa_K = D_A = D_M = 2j\rho(1 - \rho) > 0 \quad (92)$$

at every interior filling, exact pair ODLRO, and a fixed-number gap closing as L^{-2} . It is thus an explicit gapless frustration-free parent with a rigorous zero-frequency Meissner/superfluid weight. The swap backbone also protects a positive transverse floor in multiblock extensions. The next independent problem is not conceptual identification but response-stable transfer to a specified finite-coupling microscopic realization.

A Finite-volume perturbation formula

For completeness, write

$$H_L(a) = H_L(0) + aJ_L + \frac{a^2}{2}T_L + O(a^3). \quad (93)$$

With respect to $P_L \oplus Q_L$ and under $P_L J_L P_L = 0$, the Feshbach map at energies $O(a^2)$ is

$$\frac{a^2}{2}P_L T_L P_L - a^2 P_L J_L Q_L (Q_L \hat{H}_L Q_L)^{-1} Q_L J_L P_L + o(a^2). \quad (94)$$

Factoring $a^2/2$ gives (5). Thus the analytic second-order branches emerging from a degenerate ground cluster are the eigenvalues of $\mathcal{C}_L$. The lowest unresolved ground-energy curvature is $\lambda_{\min}(\mathcal{C}_L)$, while a prepared state or density matrix evaluates the corresponding expectation. These are different physical selection rules when the cluster is degenerate.

B Graph combinatorics

The subset average used in Theorem 7.3 follows from

$$\mathbb{P}(x \in S) = \frac{n}{N}, \quad (95)$$

$$\mathbb{P}(x, y \in S) = \frac{n(n-1)}{N(N-1)}, \quad x \neq y. \quad (96)$$

For zero-mean f, g ,

$$\mathbb{E} \left[\sum_{x \in S} f_x \sum_{y \in S} g_y \right] = \frac{n}{N} \sum_x f_x g_x + \frac{n(n-1)}{N(N-1)} \sum_{x \neq y} f_x g_y \quad (97)$$

$$= \frac{n(N-n)}{N(N-1)} \sum_x f_x g_x. \quad (98)$$

The same counting gives the probability that exactly one endpoint of an edge is occupied:

$$2 \frac{n}{N} \frac{N-n}{N-1} = \gamma_{N,n}. \quad (99)$$

C Normalization conventions

The electronic-link convention uses the Peierls phase of a single electron and the response density $V^{-1}\partial_A^2 E$. A pair transported across a bond acquires phase $Q = 2A$. Therefore

$$\frac{1}{V} \frac{\partial^2 E}{\partial Q^2} = \frac{1}{4V} \frac{\partial^2 E}{\partial A^2}. \quad (100)$$

This accounts for the factor of four between (71) and (72). Electric charge factors, optical π factors, and lattice-spacing tensors can be restored uniformly after choosing a material convention.

D Reproducibility

Three deterministic scripts accompany the paper.

The script `verify_dynamical_response.py` checks:

1. random finite-dimensional positive-square families against the state-space Kubo formula and the target-space identity (31);
2. the quantitative bounds (24)–(25) for random positive matrix-valued spectral weights;
3. the soft-target counterexample (85)–(87);
4. numerical convergence of the free-gas transverse cancellation.

The script `verify_graph_hodge_meissner.py` checks:

1. the unique Dicke/AGP zero state and its pair-density matrix;
2. the site-function identities (58)–(59);
3. bondwise gauge covariance;
4. the current and stress identities (62)–(63);
5. the static Hodge formula and finite-frequency graph filter;
6. exact cancellation for gradients and exact survival for lattice-transverse profiles;
7. the L^{-2} gap formula, including a multipair half-filled rectangular torus, and an operator-level multiblock floor with nontrivial residual rows.

The script `independent_fock_check.py` checks the fermionic layer before seniority-zero compression. It constructs Jordan–Wigner Fock matrices on a path, triangle, and square and verifies:

1. the complete site swaps and their Peierls lifts are Hermitian involutions;
2. the even-sector kernels have the claimed fermionic Dicke signs and odd sectors have no zero modes;
3. full-Fock gauge covariance and the charge-two seniority-zero compression;
4. current and stress derivatives against finite differences; and
5. the full-Fock static Hodge formula and finite-frequency transverse identity.

The retained outputs and SHA-256 hashes are included in the consolidated archive.

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